



BITS Pilani
DIGITAL

READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 8

Bond Portfolio Management (Part 2)



STUDY NOTES • EDITION 1.0 • WEEK 8

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How to use these notes



These notes cover the whole of Week 8: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A runs the pricing equation backwards to find yield, and asks what makes yields move. Part B measures how violently a bond responds, and compresses the answer into one number. Part C uses that number to build a portfolio that survives rate movements in either direction. Read the parts in order the first time; each assumes the last.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

Contents



GETTING ORIENTED

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A Yield in depth

1	Why the price paid decides the return
2	Computing yield, and the zero coupon case
3	Marking to market and the forces that move yields

B Volatility and duration

4	Four properties of bond price volatility
5	Duration: the balance point
6	From percentage to rupees

C Immunisation and the layers of yield

7	The two opposing effects
8	Portfolio duration and immunisation
9	One number, many names

CONSOLIDATION

Module summary, by learning objective

Glossary and self-check

Module overview



Week 7 built the bond pricing equation and ran it in one direction — rate in, price out. This module runs it the other way and then asks sharper questions of it: how much does price move when yields move, how do we measure that in a single number, and how can a portfolio be built so the movements cancel?

One thread runs through all nine topics: **duration is the number that makes bond risk manageable**. It replaces maturity and coupon with one figure, converts a yield change into a rupee loss, and — set equal to an investment horizon — turns two opposing risks into a strategy.

Learning objectives

By the end of this module, you will be able to:

- 1 **Explain** why the price paid determines return, and compute yield to maturity including the zero coupon case.
- 2 **Describe** the forces that move yields and the four properties of bond price volatility.
- 3 **Compute** Macaulay and modified duration and convert a yield change into a rupee impact.
- 4 **Apply** portfolio duration to immunise a liability, and decompose a yield into its layers.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	Why the price paid decides the return	A · Yield	L28 V1	LO 1
2	Computing yield, and the zero coupon case	A · Yield	L28 V2	LO 1
3	Marking to market and the forces that move yields	A · Yield	L29 V1	LO 2
4	Four properties of bond price volatility	B · Volatility	L29 V2	LO 2
5	Duration: the balance point	B · Duration	L29 V3	LO 3
6	From percentage to rupees	B · Duration	L29 V3	LO 3
7	The two opposing effects	C · Immunisation	L30 V1	LO 4

Topic	What it covers	Part	Source	Serves
8	Portfolio duration and immunisation	C · Immunisation	L30 V2	LO 4
9	One number, many names	C · Yield layers	L31 V1	LO 4

Key terms at a glance



Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Yield to maturity

The true economic return on a bond at its current market price.

Zero coupon bond

A bond paying no coupons, whose entire return is the gap between price and face value.

Marking to market

Updating a holding to its current market price rather than the price originally paid.

Contagion

The rapid spread of market stress from one issuer to others.

Duration

The present-value-weighted average time to receive a bond's cash flows; its centre of gravity.

Modified duration

Macaulay duration divided by $(1 + \text{yield})$; measures price sensitivity to yield changes.

Price effect

The change in a bond's market value when yields move.

Immunisation

Setting portfolio duration equal to the investment horizon so the two effects offset.

Portfolio duration

The weighted average duration of all holdings.

Term premium

The extra yield demanded for lending over a longer period.

Liquidity premium

The extra yield demanded when a bond is hard to sell quickly.

Coupon rate

The bond's contractual promise, fixed at issue and unchanging.

Treasury STRIPS

US government bonds whose coupons have been stripped and sold as separate zero coupon instruments.

Basis point

One hundredth of a percentage point; 50 basis points is 0.5 percentage points.

Convexity

The curvature of the price-yield relationship, which makes gains exceed losses for equal moves.

Macaulay duration

Duration expressed in years, measuring the timing of cash flows.

Dollar duration

The rupee value change implied by modified duration for a given yield move.

Reinvestment effect

The change in return earned on reinvested coupons when yields move.

Investment horizon

The time remaining until a liability falls due.

Term structure

The pattern of yields across maturities for the same issuer; the yield curve.

Credit spread

The gap between a corporate bond's yield and a government security of the same maturity.

Parallel shift

A movement in the risk-free rate that affects every bond in the market.

PART A • YIELD IN DEPTH

Running the equation backwards

1 Why the Price Paid Decides the Return

PART A • YIELD IN DEPTH • TOPIC 1 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Three investors buy the same bond at three prices and earn three different returns. The bond never changed — only what they paid for it.

Why this matters

Week 7 ran the pricing equation forwards: rate in, price out. This week runs it backwards, because in real markets the price is what you observe and the yield is what you must find.

The used car

Three friends buy the same used car. One pays ₹5 lakh, one negotiates down to ₹4.5 lakh, and one desperately needs a car and pays ₹5.5 lakh. Three years later all three sell it for ₹5 lakh. The car did not change and the resale value did not change — yet the returns differ entirely, because of the price paid.

A bond behaves the same way. The coupon is the fixed service you receive each year; the face value is the resale value. What determines your actual return is the price you pay today — and that actual return is the yield to maturity.

Promise and reality

The **coupon rate** is the bond's *promise*. The **yield to maturity** is the bond's *reality*. An 8% coupon bond pays ₹8 per year on ₹100 of face value whether it trades at ₹80, ₹100 or ₹120 — but the return you earn depends entirely on what you paid.

Flipping the equation

In the real world we do not choose a bond's price. The market sets it every moment of the trading day, so price is not something we calculate — it is something we observe. And if the price is known, the coupon is fixed by contract, and the face value and maturity are fixed too, then only one unknown remains in the pricing equation: the yield to maturity.

Last week the discount rate was the input and price the output. This week the price is the input and the yield the output. Same equation, run backwards.

ONE BOND, THREE BUYERS, THREE YIELDS

	At par	At a discount	At a premium
Price paid	₹100	₹95	₹105
Coupon rate	8%	8%	8%
Cash received	₹8, ₹8, ₹108	₹8, ₹8, ₹108	₹8, ₹8, ₹108
Capital gain or loss	Nil	Gain of ₹5	Loss of ₹5
Yield to maturity	8.00%	10.01%	6.13%
Versus coupon rate	Equal	Higher	Lower

Yield to maturity combines **two sources of return**: the periodic coupon payments, and the capital gain or loss booked at redemption. Only the second one varies here.

Figure 1.1 A three-year, 8% coupon, ₹100 face bond bought at three prices. Nothing about the bond differs.

KEY IDEA

Buy below face value and the capital gain pushes YTM above the coupon rate. Buy above face value and the capital loss pulls it below. Buy exactly at face and there is no gain or loss, so the two rates coincide. That relationship is arithmetic, not a market tendency.

EXAM POINTER

Distinguish coupon rate from yield to maturity in one sentence each — promise against reality — then state the three-way rule. If given a price relative to face value, you should be able to say immediately whether YTM is above, below or equal to the coupon, without computing anything.

QUICK RECAP • TOPIC 1

- The price paid determines the return; the bond itself is unchanged.
- The coupon rate is the promise; yield to maturity is the reality.
- YTM combines coupon income and the capital gain or loss at redemption.
- Bought at ₹95, ₹100 and ₹105, a three-year 8% bond yields 10.01%, 8.00% and 6.13%.
- In real markets price is observed and yield is the unknown — the equation runs backwards.

2 Computing Yield, and the Zero Coupon Case

PART A • YIELD IN DEPTH • TOPIC 2 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Yield to maturity cannot be solved by hand — except for the one bond that has no coupons at all.

Why this matters

The spreadsheet functions here are used throughout the rest of the week. The zero coupon case matters far beyond its own sake, because it is the purest illustration of what a yield actually is.

Two spreadsheet routes

=YIELD is used when the settlement and maturity *dates* are available. Its inputs are the settlement date, the maturity date, the annual coupon rate, the price paid, the redemption value and the frequency of interest payments per year.

=RATE is used when only the *number of periods* is known. Its inputs are the number of periods, the payment per period, the present value and the future value. The present value must be entered as a **negative number**, because it is a cash outflow — the money paid to buy the bond.

For the bond bought at ₹95, both return approximately **10.01%**. They are two routes to one answer.

Zero coupon bonds: the purest test case

A zero coupon bond pays no coupons at all. The investor buys at a deep discount to face value and receives the full face value at maturity, with nothing in between. The entire return is the difference between what you pay and what you receive.

This makes it the cleanest illustration of yield to maturity, because there is no coupon to confuse the picture. The coupon rate is literally zero, and yet the bond offers a positive return — and that return is the YTM. If someone told you only the coupon rate, you would think the bond pays nothing at all.

The equation simplifies too. With no coupon payments every intermediate term vanishes, leaving a single cash flow, so YTM can be solved **directly** — no trial and error and no solver needed.

Where they come from

Zero coupon bonds are issued by governments and created by financial institutions. In India the RBI issues them, and they appear in certain infrastructure financing and deep discount instruments. In the United States they are called **Treasury STRIPS** — the coupons of a regular government bond are literally stripped away and sold separately, so each individual cash flow becomes its own zero coupon instrument.

One property to carry into Topic 5: **the duration of a zero coupon bond equals its maturity exactly.**

THREE FUNCTIONS, ONE RELATIONSHIP

Function	Use it when	Note	Result on the ₹95 bond
=YIELD(settlement, maturity, rate, price, redemption, frequency)	Settlement and maturity dates are known	The standard route for a dated bond	10.01%
=RATE(nper, pmt, pv, fv)	Only the number of periods is known	Enter pv as a negative — it is a cash outflow	The same rate
Direct formula	The bond is zero coupon	One cash flow only, so no iteration is needed	Solvable by hand

For every bond with coupons, YTM requires **iteration** — you cannot rearrange the equation to isolate it. The zero coupon bond is the single exception.

Figure 2.1 Computing yield. The zero coupon case is the only one you can solve directly.

COMMON PITFALL

Entering the price as a positive number in =RATE. The function models a cash flow sequence, so the purchase must be negative and the redemption positive. Get the sign wrong and the function returns an error or a nonsensical rate.

EXAM POINTER

Name both functions with what distinguishes them — dates against periods — and flag the negative present value. For zero coupon bonds, state that the coupon rate is zero while the yield is positive, and explain why: the entire return is the capital gain from discount to face value.

QUICK RECAP • TOPIC 2

- =YIELD uses settlement and maturity dates; =RATE uses the number of periods.
- In =RATE the present value must be negative, because buying is a cash outflow.
- A zero coupon bond pays nothing until maturity; the whole return is the price gap.
- Its coupon rate is zero but its yield is positive — the cleanest illustration of what YTM means.
- With one cash flow, YTM solves directly without iteration.

3 Marking to Market, and the Forces That Move Yields

PART A • YIELD IN DEPTH • TOPIC 3 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

A bond fund can lose ₹300 crore in a single day with no default, no missed coupon and no trade. Only the market rate changed.

Why this matters

Week 7 established that price and yield move inversely. This topic asks the next question: what actually makes yields move in the first place?

Marking to market

A manager arrives at nine in the morning with a portfolio worth ₹10,000 crore. By half past three it is worth ₹9,700 crore. No company defaulted, no coupon was missed, no fraud occurred. Nothing happened to the bonds. The only thing that changed is the market interest rate.

It works like a house. On the day you sign the documents you own it, but from the next day its market value begins to change — a metro station is announced nearby and it rises; a factory opens next door and it falls. You have done nothing, and the house is unchanged, but the environment around it has.

Marking to market means updating a holding to its current market price rather than the price originally paid. The same principle drives a mutual fund's daily net asset value. Even if you never intend to sell, the bond still has a market value today, and a different one tomorrow.

Five forces that move yields

Monetary policy. When the RBI raises the repo rate, borrowing costs rise and yields across maturities generally follow. Between April 2022 and February 2023 the RBI raised the repo rate by 250 basis points, and the 10-year Government of India yield moved by more than 130 basis points.

Inflation and expected inflation. The market can react before the RBI acts. If inflation prints at 7.4% when 6.8% was expected, yields may rise within minutes — because investors lend today for fixed future payments, and higher inflation erodes the purchasing power of those payments.

Credit events and contagion. In September 2018 IL&FS defaulted despite holding the highest credit rating. Corporate bond yields rose while government yields stayed lower, and credit spreads widened sharply. Even financially strong companies saw prices fall — that rapid spread of stress is contagion.

Global spillover. When the US Federal Reserve raises rates, foreign investors may shift from Indian bonds into US Treasuries, and the selling pressure raises Indian yields.

Supply and demand. The government borrows to finance its fiscal deficit, and a large borrowing programme means the market must absorb heavy new supply. If there are not enough buyers at existing yields, yields must rise. Conversely, RBI open market purchases raise demand and push yields down.

ONE BOND, FIVE YIELDS • ₹100 FACE, 8% COUPON, FIVE YEARS

Yield to maturity	Price	Zone	Change from par
6%	₹108.42	Premium	+8.42
7%	₹104.10	Premium	+4.10
8%	₹100.00	Par	—
9%	₹96.11	Discount	-3.89
10%	₹92.42	Discount	-7.58

Three things to notice. The relationship is **not linear** — successive one-point rises cost 4.32, 4.10, 3.89 and 3.69. Gains and losses are **not symmetrical**. And a two-point move is **financially significant**, worth roughly 7 to 8% of value.

Figure 3.1 The price-yield relationship, computed for one bond. The steps shrink as yields rise.

REAL-WORLD EXAMPLE

These five forces do not operate separately. They may reinforce one another or pull in opposite directions — a repo cut may be offset by rising inflation expectations, and a flight to safety may lower government yields while raising corporate ones at the same moment.

EXAM POINTER

Name all five forces with one concrete example each; the Indian ones — the 2022 to 2023 repo cycle and the 2018 IL&FS default — carry more weight than generic statements. Define marking to market precisely: current market price, not the price originally paid.

QUICK RECAP • TOPIC 3

- Marking to market values a holding at today's price, not the price paid.
- A portfolio can lose value with no default and no trade — only rates changed.
- Five forces move yields: monetary policy, inflation, credit events, global spillover, supply and demand.
- The price-yield relationship is non-linear and asymmetric.
- A two-percentage-point yield move is worth roughly 7 to 8% of a five-year bond's value.

PART B • VOLATILITY AND DURATION

How hard does the price actually swing?

4 Four Properties of Bond Price Volatility

PART B • VOLATILITY AND DURATION • TOPIC 4 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

Direction is not enough. A risk manager needs magnitude — and four properties determine how hard any given bond will swing.

Why this matters

These four properties are what duration compresses into a single number in the next topic. Understanding them separately first is what makes duration intuitive rather than magical.

Property 1 • prices and yields move in opposite directions

A direct consequence of discounting, and the first rule of fixed income. Established in Week 7; restated here because the other three build on it.

Property 2 • the movement is asymmetric

For a given bond, a fall in yield produces a *larger* price rise than the price fall caused by an equal rise in yield. Take a 10-year 8% coupon bond priced at par and move the yield 200 basis points each way: at 6% the price is ₹114.72, a gain of 14.72%; at 10% it is ₹87.71, a loss of only 12.29%.

The gain exceeds the loss because the price-yield relationship is **curved** rather than linear. That curvature is called *convexity*. As yields fall, the present value of distant cash flows increases at an increasing rate; as yields rise, present values fall but the price can never go below zero.

Property 3 • longer maturities are more sensitive

Four bonds, each with an 8% coupon priced at par, with the yield moving from 8% to 7% and then to 9%. The 3-year bond swings ₹5.16; the 10-year ₹13.44; the 20-year ₹19.72; the 30-year ₹22.68.

Same coupon, same yield movement — yet the 30-year bond is about **4.4 times** as volatile as the 3-year. It receives a larger proportion of its value from cash flows far in the future, and distant cash flows are discounted over more years, so a small change in yield is applied repeatedly over a longer period.

Property 4 • lower coupons are more sensitive

Three bonds, each 10 years to maturity at a starting yield of 8%, when the yield falls by one percentage point. The 12% coupon bond rises 6.53%; the 4% coupon bond 7.89%; the zero coupon bond **9.75%**.

A high-coupon bond pays out much of its cash early, and those early payments anchor its price against changes in the discount rate. A zero coupon bond concentrates its entire value in a single payment at maturity, making it the most price-sensitive of all.

MATURITY AND COUPON • THE TWO LEVERS THAT SET VOLATILITY

**MATURITY EFFECT • 8% COUPON, YIELD 7%
AGAINST 9%****3 YEARS**

Swing of ₹5.16

10 YEARS

Swing of ₹13.44

20 YEARS

Swing of ₹19.72

30 YEARS

Swing of ₹22.68 — about 4.4× the 3-year

**COUPON EFFECT • 10 YEARS, YIELD 8% FALLING
TO 7%****12% COUPON**

₹126.84 → ₹135.12, up 6.53%

4% COUPON

₹73.16 → ₹78.93, up 7.89%

ZERO COUPON

₹46.32 → ₹50.83, up 9.75%

WHY

Early cash anchors the price; no early cash means none

Both effects point the same way: **the further away the cash flows, the harder the price swings**. Duration in Topic 5 is precisely the number that captures how far away they are.

Figure 4.1 The two levers. Longer maturity and lower coupon both push cash flows into the future.

KEY IDEA

Maturity and coupon are not two separate causes. They are two ways of doing the same thing — shifting the *average timing* of the bond's cash flows further out. That single insight is what allows one number to replace both.

EXAM POINTER

Name all four properties. Property 2 requires the word *convexity* and the reason — the curve bends, and price cannot fall below zero. For properties 3 and 4, give the direction and the mechanism: distant cash flows are discounted over more periods, and early coupons anchor the price.

QUICK RECAP • TOPIC 4

- Prices and yields move in opposite directions — the first rule of fixed income.
- The movement is asymmetric: a 200bp fall gains 14.72% where a 200bp rise loses 12.29%. That curvature is convexity.
- Longer maturities swing harder; a 30-year bond is about 4.4 times as volatile as a 3-year.
- Lower coupons swing harder; the zero coupon bond is the most sensitive of all.
- Both effects work by pushing the average timing of cash flows further into the future.

5 Duration: The Balance Point

PART B • VOLATILITY AND DURATION • TOPIC 5 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

One number replaces both maturity and coupon — the weighted average time at which you actually receive a bond's cash flows.

Why this matters

Duration is the central measure of the week and the input to immunisation in Topic 8. It is also the number whose misuse contributed to a bank failure in 2023.

The plank and the weights

Picture a bond's cash flows as weights placed along a plank at different distances. Each coupon is a small weight at the end of year one, year two, year three and so on; the principal repayment is a large weight at the maturity date. **Duration is the balance point of that plank** — the weighted average time at which you receive the present-value-weighted cash flows. It is measured in years.

Duration is not maturity

Maturity tells us when the life of a bond ends. Duration tells us how strongly its price reacts when interest rates change. When coupons are large and frequent they pull the balance point closer to the present, so duration is *shorter* than maturity. When there are no coupons at all, the only cash flow arrives at maturity — so a zero coupon bond's duration equals its maturity exactly.

Three steps from duration to a price estimate

One. Calculate the Macaulay duration — the weighted average time at which the bondholder receives all cash flows. In Excel, =DURATION.

Two. Convert to **modified duration** by dividing Macaulay duration by $(1 + \text{yield to maturity})$. In Excel, =MDURATION.

Three. Estimate the percentage price change: **approximate % change** $\approx$ **-modified duration** $\times$ **change in yield**. A bond with modified duration of 8 years, when rates rise one percentage point, is expected to fall by $-8 \times 0.01 = 8\%$.

This is an approximation, because the true relationship is curved. The actual rise when yields fall is usually slightly greater than estimated, and the actual fall when yields rise slightly smaller — that is convexity again, working in the holder's favour.

Four properties of duration

A **zero coupon bond's duration equals its maturity**, since the entire weight sits at the maturity date. Duration **increases with maturity but at a decreasing rate** — a 30-year bond's duration is not twice a 15-year bond's. Duration **decreases as the coupon rises**, because more cash arrives early. And duration **decreases as the yield rises**, because higher yields shrink distant present values more than near ones.

COMPUTING MACAULAY DURATION • ₹100 FACE, 8% COUPON, FIVE YEARS, PRICED AT PAR

Year	Cash flow	Present value at 8%	Weight	Time × weight
1	₹8	₹7.41	0.0741	0.0741
2	₹8	₹6.86	0.0686	0.1372
3	₹8	₹6.35	0.0635	0.1905
4	₹8	₹5.88	0.0588	0.2352
5	₹108	₹73.50	0.7350	3.6750
Total	—	₹100.00	1.0000	4.312

The bond matures in 5 years but its duration is 4.31. The coupons pull the centre of gravity forward. Modified duration = $4.31 \div 1.08 = 3.99$, so a one-point yield rise cuts the price by about 3.99%.

Figure 5.1 The duration calculation. Note that year five contributes 73.5% of the total weight.

KEY IDEA

Duration is not a forecast of when you get your money back in cash terms. It is the *present-value* weighted average timing — which is exactly the quantity that determines how much a change in the discount rate moves the price.

EXAM POINTER

Build the table with all five columns and show the total weight summing to 1.0000 — that is a useful self-check. Convert to modified duration and apply the price-change approximation. State that it is an approximation and name convexity as the reason.

QUICK RECAP • TOPIC 5

- Duration is the present-value-weighted average time at which cash flows arrive, measured in years.
- It is not maturity: coupons pull the balance point forward, so duration is shorter.
- A five-year 8% bond at par has a Macaulay duration of 4.31 years.
- Modified duration = $\text{Macaulay} \div (1 + \text{YTM})$; here $4.31 \div 1.08 = 3.99$.
- Approximate % price change $\approx -\text{modified duration} \times \text{change in yield}$.

6 From Percentage to Rupees

PART B • VOLATILITY AND DURATION • TOPIC 6 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

A percentage is an abstraction. Applied to a portfolio, modified duration turns a 25 basis point move into a number the risk committee can act on.

Why this matters

This is where duration stops being a classroom quantity. The 2023 case study shows what the arithmetic looks like when the portfolio is large enough and the rate move sustained enough.

Dollar duration

For a portfolio manager the rupee impact matters more than the percentage. **Dollar duration** converts the percentage change into money by applying it to the portfolio value:

Rupee change $\approx$ **–modified duration** $\times$ **change in yield** $\times$ **portfolio value**

Which is how a move of just 25 basis points can translate into a loss of several crore on a large book. The percentage looks small; the rupee figure does not.

Silicon Valley Bank, March 2023

SVB held approximately **\$91 billion** in securities classified as held to maturity — mainly long-term mortgage-backed securities and US Treasuries. The estimated portfolio duration was about 6 to 7 years.

Across 2022 and 2023 the US Federal Reserve raised rates by a total of **525 basis points**. Applying the duration estimate: $-6.5 \times 0.0525 \times \$91\text{bn} \approx$ **–\$31 billion**.

The actual unrealised loss reached about **\$17 billion** — lower than the estimate, because of coupon income and the gradual repayment of principal on the mortgage-backed securities. Even so, \$17 billion exceeded the bank's total equity.

What the case actually teaches

The duration calculation was not precise, and it did not need to be. It correctly indicated the **order of magnitude** of the danger — tens of billions against an equity base far smaller than that. A rough number computed early would have been worth far more than a precise one computed too late.

Note also what did *not* happen: no borrower defaulted. Every security paid as promised. The loss was entirely a rate-driven revaluation, which stayed invisible while the bonds were held and became real the moment they had to be sold.

THE SVB ARITHMETIC • A ROUGH ESTIMATE THAT WAS RIGHT ENOUGH

Step	Input or working	Result
Portfolio	Held-to-maturity securities, mainly long-term MBS and Treasuries	\$91 billion
Estimated duration	Long-dated, low-coupon holdings	About 6 to 7 years
Rate move	US Federal Reserve, 2022 into 2023	525 basis points
Duration estimate	$-6.5 \times 0.0525 \times \91bn	$\approx -\$31$ billion
Actual unrealised loss	Lower, due to coupon income and MBS principal repayment	About $-\$17$ billion
Against the bank's equity	The loss exceeded total equity	Failure

*No borrower defaulted. Every security paid as promised. The loss was **entirely rate-driven** — invisible while held, and real the moment the bonds had to be sold.*

Figure 6.1 Duration applied at scale. The estimate overshot, and still gave the right warning.

COMMON PITFALL

Assuming that a bond classified as held to maturity is insulated from rate risk. The accounting treatment hides the loss; it does not remove it. The moment a forced sale becomes necessary, the paper loss becomes a realised one — which is precisely what happened to SVB.

EXAM POINTER

Write the dollar duration formula with all three terms and apply it. If asked about SVB, give the calculation, note that the estimate exceeded the actual loss, and explain why — coupon income and principal repayment. The examinable insight is that the estimate was directionally right and that was enough.

QUICK RECAP • TOPIC 6

- Rupee change $\approx$ $-\text{modified duration} \times \text{change in yield} \times \text{portfolio value}$.
- A 25 basis point move can cost several crore on a large portfolio.
- SVB held \$91bn at a duration of about 6.5 years when rates rose 525 basis points.
- The duration estimate gave $-\$31\text{bn}$; the actual unrealised loss was about $-\$17\text{bn}$.
- No borrower defaulted — the loss was entirely a rate-driven revaluation.

PART C • IMMUNISATION AND THE LAYERS OF YIELD

Turning two risks into a strategy

7 The Two Opposing Effects

PART C • IMMUNISATION AND THE LAYERS OF YIELD • TOPIC 7 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Rising rates cut your price and lift your coupon reinvestment. Falling rates do the reverse. The two always pull in opposite directions.

Why this matters

Week 7 named these two risks. This topic works them through numerically, which is what reveals that they can be made to cancel.

The provident fund problem

You manage a provident fund with a single known liability: ₹100 crore payable to retirees exactly five years from today. Your job is to invest now so that whatever happens to interest rates, you have exactly ₹100 crore when the liability falls due.

The complication is that a five-year coupon-paying bond pays coupons along the way, and those coupons must be reinvested. If rates fall, they are reinvested at lower rates and the accumulated total may fall short. If rates rise, the coupons earn more — but the market value of the remaining bond position falls.

Working it through

Rahul buys an ABC Ltd bond: ₹100 face value, 8% annual coupon, five years to maturity, purchased at par so the yield at purchase is 8%. He plans to hold for three years and then sell, at which point the bond will have two years remaining. He receives ₹8 at the end of each year and reinvests each coupon at the prevailing rate.

If rates stay at 8% throughout, everything works as expected. But the moment rates change, two things happen at once — and they pull in opposite directions.

Why they do not cancel here

The two effects consistently move in opposite directions, but they are not necessarily *equal in size*. In this example the capital loss of ₹3.47 is far larger than the reinvestment gain of 51 paise, so Rahul is worse off overall.

This happens because his three-year holding period is **shorter** than the effective timing of the bond's remaining cash flows — recall that this bond's duration is 4.31 years. When the horizon is shorter than the duration, the price effect dominates.

Which raises the obvious question: is there a holding period at which the two exactly cancel? There is — and it is the duration of the bond.

THE SAME BOND, TWO RATE SCENARIOS • OPPOSITE TRIGGERS, OPPOSITE SIGNS

	Rates rise from 8% to 10%	Rates fall from 8% to 6%
Selling price after 3 years, 2 remaining	$8 \div 1.10 + 108 \div 1.10^2 = \text{₹}96.53$	$8 \div 1.06 + 108 \div 1.06^2 = \text{₹}103.67$
Capital effect	Loss of ₹3.47 — unfavourable	Gain of ₹3.67 — favourable
Coupons reinvested	$(8 \times 1.10^2) + (8 \times 1.10) + 8 = \text{₹}26.48$	$(8 \times 1.06^2) + (8 \times 1.06) + 8 = \text{₹}25.47$
Against ₹25.97 at 8%	Gain of 51 paise — favourable	Loss of about 50 paise — unfavourable
Net outcome	The ₹3.47 loss dwarfs the 51 paise gain	The ₹3.67 gain dominates

ABC Ltd did nothing wrong. Every coupon was paid on time. The price moved purely because the market now requires a **different return**.

Figure 7.1 The two effects worked through. Note they are opposite in sign but unequal in size.

KEY IDEA

The price effect and the reinvestment effect are not two separate problems to be solved. They are one problem with two signs — and because the signs are always opposite, the right holding period makes them cancel rather than compound.

EXAM POINTER

Compute both effects for both scenarios and show the workings. The high-value observation is *why* they fail to cancel here: the three-year horizon is shorter than the 4.31-year duration, so the price effect dominates. That sentence sets up Topic 8 and is usually worth a mark on its own.

QUICK RECAP • TOPIC 7

- A single known future liability creates the problem immunisation solves.
- Rising rates cut the selling price but raise coupon reinvestment income.
- Falling rates raise the price but cut reinvestment income.
- For Rahul, a ₹3.47 capital loss dwarfs a 51 paise reinvestment gain.
- They fail to cancel because his three-year horizon is shorter than the bond's 4.31-year duration.

8 Portfolio Duration and Immunisation

PART C • IMMUNISATION AND THE LAYERS OF YIELD • TOPIC 8 OF 9 • SERVES LEARNING OBJECTIVE 4

IN ONE LINE

Set the portfolio's duration equal to the investment horizon and the two effects offset. That is immunisation.

Why this matters

This is the practical payoff of the entire module — the technique that turns a pair of risks into a strategy, and the reason duration is computed at all.

Portfolio duration

Managers rarely hold a single bond. The duration of a bond portfolio is the **weighted average duration of all its holdings**, with each weight being that bond's share of total portfolio value.

Bond A has a duration of 4 years and is 40% of the portfolio, contributing $0.40 \times 4 = 1.6$ years. Bond B has a duration of 10 years and is 60%, contributing $0.60 \times 10 = 6.0$ years. Portfolio duration is **7.6 years** — the portfolio behaves as though the weighted average timing of its cash flows is 7.6 years away.

The principle

Immunisation means constructing a portfolio whose Macaulay duration equals the investment horizon, or the time remaining until a future liability falls due.

Think of it as synchronising two clocks. The first represents the timing of cash flows received from the portfolio; the second represents the time at which the money is required. Immunisation keeps the two aligned. If a liability is due in five years, the manager builds a portfolio with a Macaulay duration of five years.

When duration matches the horizon, the gain from one effect approximately offsets the loss from the other for small, parallel changes in yields — so the portfolio value at the end of the horizon remains close to the amount required.

Where it is used, and what it really teaches

Common applications include a child's university fees seven years away, pension payments due in twelve years, or an insurance claim expected in five.

The wider lesson matters beyond the technique. Every coupon-paying bond carries both interest rate risk and reinvestment risk, and the two move in opposite directions. **The objective of bond portfolio management is not always to eliminate risk completely — it is often to balance one form of risk against the other.**

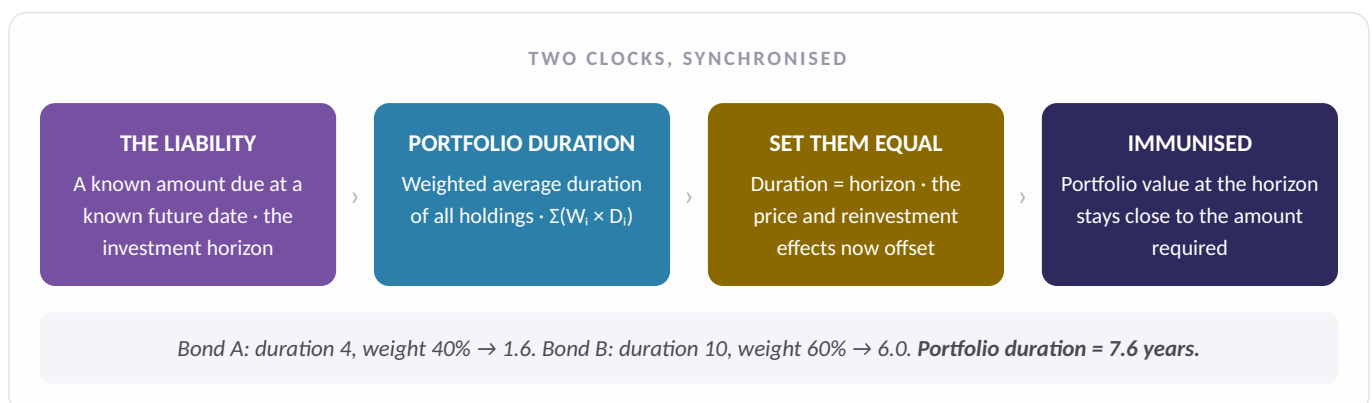


Figure 8.1 Immunisation. The technique works for small, parallel shifts in the yield curve.

KEY IDEA

Immunisation does not remove risk. It *arranges* two risks so that they cancel — which is a different and more achievable goal, and one available only because the two effects have opposite triggers.

COMMON PITFALL

Matching *maturity* to the horizon rather than duration. A five-year bond does not immunise a five-year liability — its duration is only 4.31 years, so the reinvestment effect would dominate. Duration is the correct clock, not maturity.

EXAM POINTER

Compute portfolio duration as a weighted average, showing each product. State the immunisation condition in one line — portfolio duration equals the investment horizon — and name the assumption it rests on: small, parallel yield changes.

QUICK RECAP • TOPIC 8

- Portfolio duration is the weighted average duration of the holdings: $\sum (W_i \times D_i)$.
- Durations of 4 and 10 at weights 40% and 60% give a portfolio duration of 7.6 years.
- Immunisation sets portfolio duration equal to the investment horizon.
- The price and reinvestment effects then offset for small, parallel yield changes.
- Match duration to the horizon, never maturity — they are not the same number.

9 One Number, Many Names

PART C • IMMUNISATION AND THE LAYERS OF YIELD • TOPIC 9 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

The discount rate, the yield to maturity and the required rate of return are the same number — and it is built from four separate layers.

Why this matters

The closing topic of the module, and the one that resolves two weeks of vocabulary into a single idea before decomposing it into its components.

Six terms, one number

Across two weeks you have met the discount rate, yield to maturity, internal rate of return, required rate of return, opportunity cost of capital and market interest rate. The point rarely stated clearly enough is this: **for a given bond, at a given market price, on a given day, all six refer to the same number.** What changes is the question being asked.

Call it the *discount rate* and you are asking which rate finds the present value. Call it the *yield to maturity* and you are asking what annualised return the bond offers at its current price. Call it the *internal rate of return* and you are asking which rate makes net present value zero. Call it the *opportunity cost of capital* and you are asking what is given up by not investing elsewhere. Call it the *required rate of return* and you are asking what minimum return justifies the risk. Call it the *market interest rate* and you are asking what comparable bonds yield today.

There is no single market interest rate

When a headline says interest rates have risen, it is using the term broadly. In reality thousands of rates operate simultaneously, and every bond has its own.

Consider blood pressure. A reading is not judged against one universal number — it is assessed against the person's age, weight and history. Bonds work the same way. Each belongs to a category of maturity, credit risk and liquidity, and the market sets an appropriate rate for that category.

Two patterns explain the differences. The **term premium**: comparing 2, 5 and 10-year government securities, only maturity differs, and longer maturities yield more because lending over a longer period brings greater uncertainty. That pattern across maturities is the *term structure*, or yield curve. The **credit spread**: comparing bonds of the same maturity but different ratings, the government security sits lowest and the weakest-rated issuer highest.

Reading price volatility through the layers

When a bond's price changes, one or more layers of its yield have shifted. If **the RBI raises the repo rate**, the risk-free rate moves for every bond — a parallel shift in the yield curve affecting government and corporate, AAA and BBB alike. If **a rating agency downgrades one issuer**, only that issuer's credit spread moves; the G-Sec yield is unchanged and other corporates are unaffected. If **a market-wide credit shock hits**, as with IL&FS, spreads widen across all corporates simultaneously even for companies with strong fundamentals — and the risk-free rate may actually *fall* as investors flee to government bonds, so government prices rise while corporate prices fall, both responding to the same event.

FOUR LAYERS INSIDE A SINGLE YIELD • AN ILLUSTRATIVE 5-YEAR CORPORATE BOND

Layer	Value	What it compensates for	How it is set
Risk-free rate	7.10%	The basic time value of money	The yield on a comparable 5-year Government of India security
Credit spread	2.30%	The risk that the issuer may not pay	Assessed by rating agencies; weaker issuer, larger spread
Liquidity premium	0.50%	The difficulty of selling quickly	Harder to measure; important in the Indian corporate bond market
Other market factors	0.30%	Residual supply, demand and structural effects	Market determined
Total yield to maturity	10.20%	—	—

*A total yield is **not one decision**. It is the combined price of several different risks, which is why a yield should never be judged in isolation but always against comparable bonds.*

Figure 9.1 An illustrative decomposition. The four layers sum to the quoted yield.

KEY IDEA

Because a yield is built from layers, an unusually high one is not automatically an opportunity. It may signal a credit spread the market has widened for good reason. The useful question is always *which layer is elevated*, not whether the total looks attractive.

EXAM POINTER

State that the six names describe one number and give at least three with the question each answers. Then decompose a yield into its four layers with what each compensates for. If asked about a price move, identify *which layer* shifted — parallel shift, issuer-specific, or market-wide credit shock.

QUICK RECAP • TOPIC 9

- Discount rate, YTM, IRR, required return, opportunity cost and market rate are one number, six questions.
- There is no single market interest rate — every bond has its own, set by its category.
- The term premium explains differences across maturities; the credit spread across ratings.
- A yield decomposes into risk-free rate, credit spread, liquidity premium and other factors.
- A repo change shifts every yield; a downgrade shifts one issuer's spread; a credit shock widens all spreads at once.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Explain why price determines return and compute yield to maturity	Topics 1 and 2 • Figures 1.1, 2.1
2	Describe what moves yields and the four properties of price volatility	Topics 3 and 4 • Figures 3.1, 4.1
3	Compute duration and convert a yield change into a rupee impact	Topics 5 and 6 • Figures 5.1, 6.1
4	Immunise a liability and decompose a yield into its layers	Topics 7, 8 and 9 • Figures 7.1, 8.1, 9.1

The module in six sentences

The coupon rate is a bond's promise and the yield to maturity is its reality, so the same bond bought at ₹95, ₹100 and ₹105 yields 10.01%, 8.00% and 6.13%. Yield cannot be solved algebraically except for the zero coupon bond, whose entire return is the gap between purchase price and face value. Bond prices are marked to market daily and move under five forces — monetary policy, inflation, credit events, global spillover and supply and demand — through a relationship that is curved rather than linear, so gains from falling yields exceed losses from equal rises. Four properties govern how violently any bond responds, and duration compresses maturity and coupon into one number: the present-value-weighted average timing of the cash flows, 4.31 years for a five-year 8% bond at par. Applied to a portfolio, modified duration converts a yield change into a rupee loss, which is the arithmetic that destroyed Silicon Valley Bank without a single borrower defaulting. And because the price and reinvestment effects always pull in opposite directions, setting portfolio duration equal to the investment horizon makes them cancel — which is immunisation.

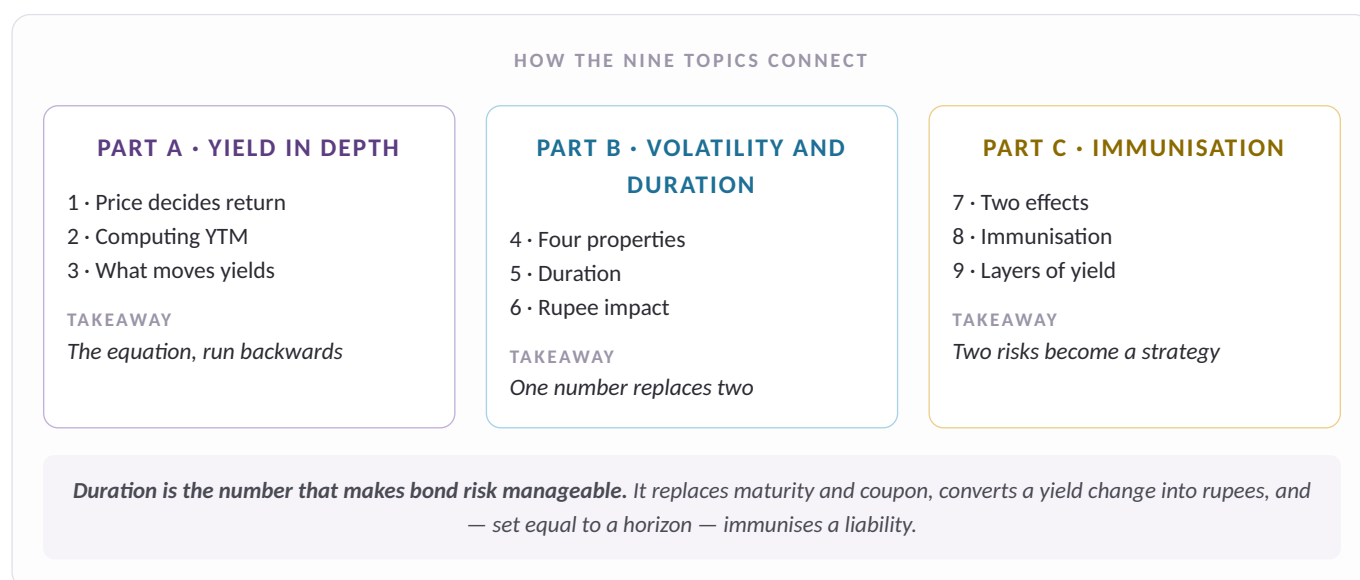


Figure M1 How the nine topics connect. Revise along the parts, not down the list.

Glossary and self-check



Term	Definition
Yield to maturity	The true economic return on a bond at its current market price.
Coupon rate	The bond's contractual promise, fixed at issue and unchanging.
Zero coupon bond	A bond paying no coupons, whose entire return is the gap between price and face value.
Treasury STRIPS	US government bonds whose coupons have been stripped and sold as separate zero coupon instruments.
Marking to market	Updating a holding to its current market price rather than the price originally paid.
Basis point	One hundredth of a percentage point; 50 basis points is 0.5 percentage points.
Contagion	The rapid spread of market stress from one issuer to others.
Convexity	The curvature of the price-yield relationship, which makes gains exceed losses for equal moves.
Duration	The present-value-weighted average time to receive a bond's cash flows; its centre of gravity.
Macaulay duration	Duration expressed in years, measuring the timing of cash flows.
Modified duration	Macaulay duration divided by $(1 + \text{yield})$; measures price sensitivity to yield changes.
Dollar duration	The rupee value change implied by modified duration for a given yield move.
Price effect	The change in a bond's market value when yields move.
Reinvestment effect	The change in return earned on reinvested coupons when yields move.
Immunisation	Setting portfolio duration equal to the investment horizon so the two effects offset.
Investment horizon	The time remaining until a liability falls due.
Portfolio duration	The weighted average duration of all holdings.
Term structure	The pattern of yields across maturities for the same issuer; the yield curve.
Term premium	The extra yield demanded for lending over a longer period.
Credit spread	The gap between a corporate bond's yield and a government security of the same maturity.
Liquidity premium	The extra yield demanded when a bond is hard to sell quickly.
Parallel shift	A movement in the risk-free rate that affects every bond in the market.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 A three-year 8% coupon bond with ₹100 face value is bought at ₹95. Is its YTM above or below 8%, and why?

Above — 10.01%. You pay ₹95 but receive ₹100 at maturity, so a ₹5 capital gain adds to the coupon income. Buying below face value always pushes YTM above the coupon rate.

Q2 A 10-year 8% bond at par moves 200 basis points each way. Why is the gain of 14.72% larger than the loss of 12.29%?

Convexity. The price-yield relationship is curved, not straight. As yields fall the present value of distant cash flows rises at an increasing rate; as yields rise, prices fall but can never go below zero. The curvature works in the holder's favour.

Q3 A five-year bond with an 8% coupon trades at par. Its Macaulay duration is 4.31 years. What is its modified duration, and what does a one-point yield rise do to the price?

Modified duration 3.99; the price falls about 3.99%. Modified = Macaulay $\div$ (1 + YTM) = 4.31 $\div$ 1.08. Then approximate % change = -modified duration $\times$ change in yield = -3.99 $\times$ 0.01.

Q4 SVB held \$91bn at a duration of about 6.5 years when rates rose 525 basis points. Estimate the loss, and say why the actual figure was smaller.

About -\$31 billion estimated; roughly -\$17 billion actual. $-6.5 \times 0.0525 \times 91 = -\31bn . The actual loss was lower because of coupon income received and gradual principal repayment on the mortgage-backed securities. Even so, \$17bn exceeded the bank's equity.

Q5 A liability falls due in five years. Why does buying a five-year bond fail to immunise it?

Because duration, not maturity, is the correct clock. A five-year 8% coupon bond has a duration of only 4.31 years, so the horizon exceeds the duration and the reinvestment effect would dominate. You need a portfolio whose *duration* is five years.

Q6 A corporate bond yields 10.20% while a comparable government security yields 7.10%. What accounts for the gap?

Credit spread, liquidity premium and other market factors. In the illustrative decomposition: 2.30% credit spread for default risk, 0.50% liquidity premium for the difficulty of selling quickly, and 0.30% residual. A high yield is not automatically an opportunity — ask which layer is elevated.